

AI Tools & Usage Documentation

Autonomous Patient Access Agent, Capabilities & Safety Architecture

Author / Role: AI Systems & Healthcare Safety Engineering	Primary Model: Google Gemini 2.5 Flash (via Google GenAI SDK)
Safety Protocol: Emergency Triage & Anti-Hallucination Guardrails	Execution Model: 17 Strongly-Typed Capability Tools

1. Clinical Safety Guardrails & Emergency Escalation

The AI assistant is strictly engineered as an **Administrative Healthcare Access Coordinator**. It is fundamentally constrained from engaging in clinical diagnosis or medical decision-making.

MANDATORY CLINICAL SAFETY BOUNDARIES:

- **NO MEDICAL DIAGNOSIS:** The AI is forbidden from diagnosing symptoms, assessing disease severity, or offering medical prognoses.
- **NO DRUG PRESCRIBING:** The AI cannot prescribe medications, suggest dosages, or recommend medication alterations.
- **ACUTE EMERGENCY PROTOCOL:** If a user expresses emergency symptoms ('severe chest pain', 'shortness of breath', 'stroke signs', 'uncontrolled bleeding'), the AI immediately terminates scheduling, advises dialing 911 / attending the nearest Emergency Department, and marks the interaction as an emergency escalation in the platform audit logs.

2. Dual-Provider Architecture & Operational Resilience

To ensure zero downtime and reliable evaluation in offline or air-gapped environments, the system implements an abstract AIProvider interface with two operational modes:

- **Primary Provider (GeminiProvider):** Utilizes the Google GenAI SDK with Gemini 2.5 Flash. Executes structured tool calling where capabilities are declared as OpenAPI-compatible function declarations.
- **Fallback Provider (FallbackProvider):** A deterministic regex- and keyword-driven semantic parser. If no GEMINI_API_KEY is provided or if network connectivity is interrupted, the fallback provider handles doctor search, availability queries, bookings, emergency escalation, and pre-visit questionnaires with 100% test passing accuracy.
- **Automatic Downgrade:** The system monitors API health; any API quota exhaustion or upstream network timeout automatically drops to the fallback provider without dropping patient sessions.

3. Controlled Capability Architecture (17 Registered Tools)

The AI model **never has direct database or SQL access**. All reads and mutations occur via 17 strongly-typed, schema-validated capabilities:

Capability Name	Domain / Scope	Functionality & Inputs
search_hospitals	Hospital Discovery	Finds approved hospitals by name, city, or postal code.
search_doctors	Doctor Discovery	Finds active doctors by medical specialty (e.g. Orthopedics, Cardiology) or name.
check_availability	Scheduling	Queries database SchedulingService for real available unbooked slots for a doctor.

lookup_patient	Patient Identity	Resolves patient profile and demographics by email or phone.
get_appointment	Appointments	Retrieves appointment details, status, and associated questionnaires.
create_appointment	Appointments	Books a real slot with two-phase EHR sync and idempotency key.
reschedule_appointment	Appointments	Atomically reschedules to a new valid slot with downstream EHR update.
cancel_appointment	Appointments	Cancels booking, releases slot back to availability pool, updates EHR.
get_questionnaire	Clinical Intake	Retrieves pre-visit questionnaire schema assigned to the appointment.
submit_questionnaire	Clinical Intake	Submits patient answers (symptoms, duration, allergies) for doctor review.
send_notification	Notifications	Dispatches multi-channel alerts (SMS, Email, In-App).
start_workflow	Workflows	Triggers automated event workflows (e.g. 24h pre-visit reminders).
get_context	Session State	Inspects conversation session state, pending slots, and user history.
update_preferences	Patient Preferences	Updates communication preferences (SMS, email, voice call).
verify_external_appointment	EHR Integration	Checks whether appointment exists in external EHR using idempotency key.
synchronize_state	EHR Integration	Synchronizes internal appointment status with downstream EHR state.
transfer_to_human	Human Escalation	Escalates complex requests or medical emergencies to human clinical staff.

4. AI Engineering Workflow & Tooling Documentation

The platform was engineered using modern agentic AI development practices and development tooling:

- **Development Environment:** Google Antigravity Agentic IDE & Python 3.12+ toolchain.
- **AI Code Generation & Verification:** Model-assisted test-driven development (TDD) utilizing Gemini 3.8 Flash to synthesize comprehensive unit and integration test fixtures for complex edge cases (e.g. concurrent race condition booking, network timeout injections, state machine invariants).
- **Structured Schema Validation:** Pydantic v2 schemas were generated and verified against the Healthcare Connector interface to ensure strict JSON serialization without runtime type errors.
- **Automated Evaluation Pipeline:** 100% test pass rate across 6 test modules verifying multi-tenancy, availability, state machine, failure recovery, and safety boundaries.